

1. Introduction

Unemployment and skill mismatch remain major challenges among students and fresh graduates. Many individuals are unsure about which career paths align with their skills. Artificial Intelligence offers an opportunity to provide intelligent, scalable, and personalized career guidance.

2. Problem Statement

Students and job seekers often lack clarity about suitable job roles and required skills. Traditional career counseling is limited in reach and accessibility. This gap leads to inefficient career decisions and reduced employment readiness.

3. Project Objective

The objective of this project is to design and develop an AI-powered career guidance agent that analyzes user skills and provides job role recommendations, skill gap analysis, and learning suggestions aligned with SDG 8.

4. SDG Alignment

This project aligns with **SDG 8: Decent Work and Economic Growth** by:

- Improving employability
 - Reducing skill mismatch
 - Supporting sustainable workforce development
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5. Proposed AI Solution

The solution is a prompt-based AI agent developed using Python. The system accepts user skills as input, processes them using AI-based reasoning, and generates career recommendations. The architecture is designed to integrate IBM Watsonx or similar AI models in future deployments.

6. Methodology

- User inputs skills or resume text
- AI analyzes skills using NLP concepts
- Job roles and skill gaps are identified
- Personalized learning recommendations are generated

7. Tools & Technologies

- Python
- Google Colab
- IBM SkillsBuild platform
- Simulated AI logic (API-ready architecture)

8. Ethical and Responsible AI Use

- No personal data storage
- Bias-aware recommendations
- Transparent and explainable outputs
- Responsible AI practices followed

9. Expected Outcomes

- Increased career awareness
- Better skill alignment
- Improved employment readiness
- Contribution to economic growth

10. Conclusion

The project demonstrates the application of Applied AI to a real-world employment challenge. By aligning with SDG 8, the solution promotes decent work opportunities and sustainable economic development.